

Personal Statement

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UK Global Talent Visa — Digital Technology (Exceptional Promise)

I am applying under the Global Talent route in Digital Technology (Exceptional Promise) as a security engineer and technical founder whose work combines cybersecurity, machine learning, blockchain compliance, and privacy-preserving verification. My application is based on independently verifiable outputs: a publicly deployed compliance platform (AMTTP), a substantial public codebase, reproducible ML benchmarks, security audit artefacts, research preprints demonstrating early academic uptake, and a UK patent filing. Together, these show original technical innovation, emerging external recognition, and credible potential to contribute meaningfully to the UK digital technology sector.

Pathway and Independent Incubation

Building on a Bachelor of Technology in Computer Science and Engineering, my earlier professional experience in cybersecurity, enterprise API integrations, logistics operations, and vendor risk exposed the practical gap that became the focus of my work: financial and digital-asset systems were becoming faster and more distributed, while compliance controls remained fragmented, reactive, and difficult to verify.

To address this, I spent my first three years in the United Kingdom focused on deep, independent research and development. Operating outside traditional institutional structures allowed me to approach this problem from first principles. During this period, I developed the mathematical and technical foundations of the platform, engineering the risk engine, smart-contract enforcement layer, and security architecture. This deliberate technical incubation culminated in the public release of my work from late 2025 onwards: live deployment, public code, audit validations, and formal IP filings. I rely not on conventional job titles, but on what I have demonstrably built, published, and had independently recognised.

What I Have Built: The AMTTP Infrastructure

AMTTP (Anti-Money Laundering Transaction Trust Protocol) is a blockchain-native compliance system designed to make risk decisions before transaction finality. Most digital-asset compliance tooling operates after settlement, when funds have already moved. AMTTP addresses that gap by combining machine-learning risk scoring, relationship analysis, sanctions screening, and policy evaluation inside the blockchain confirmation window, so a transaction can be approved, escrowed, or blocked before settlement.

Rather than listing every component of the architecture, the importance of AMTTP lies in its outcome: it is a production-style compliance platform capable of pre-settlement

decisioning, explainability, and privacy-preserving verification. It is currently deployed publicly on the Ethereum Sepolia testnet. The system is secure-by-design; through zk-NAF, it supports sanctions non-membership verification and KYC-related confirmations without unnecessary data exposure, leveraging Groth16 SNARKs. The contract suite's robustness is validated by Slither static analysis and Echidna property-based fuzzing at highest iteration depths.

Applied Machine Learning and Original Research

My innovation in machine learning directly powers the AMTTP platform. I developed and evaluated distilled student models on over 625,000 transaction samples. Rather than rely on theoretical maximums, I focused on deployability: I engineered distilled LightGBM and XGBoost models that achieve strong fraud-detection performance and robustness while remaining highly efficient on commodity CPUs, removing the dependency on specialist GPU infrastructure for commercial deployment.

Alongside the platform, I published four research preprints detailing the architecture, Blind-Spot Decomposition Theory (BSDT), Universal Detection Principle, and system stability. The value of this research is that it directly informed a working platform that is fully deployed today. Furthermore, BSDT which diagnoses classifier blind spots without retraining has already seen early academic validation. In April 2026, three of my papers were adopted as recommended readings for the postgraduate course CPE 604 by Dr Sunday Adeola Ajagbe, an independent academic with no prior supervisory relationship to me. This demonstrates that my independent research is both practically applied and externally validated.

Commercial Intent and UK Contribution

On 4 March 2026, I filed UK patent application No. 1026066039, a comprehensive filing covering the machine-learning risk scoring, blockchain compliance architecture, zero-knowledge identity workflows, and adaptive friction mechanisms I built. This IP filing signals my intent to transition AMTTP from R&D into a viable commercial infrastructure.

The United Kingdom is the optimal environment for this stage of my career. The UK's robust fintech and cyber-security ecosystems, coupled with the Financial Conduct Authority (FCA's) progressive digital-asset agenda, make dynamic compliance infrastructure commercially vital. I intend to remain in the UK long term. My focus will be on building regulated pilots, engaging with FCA-aligned compliance use cases, collaborating with UK fintechs, and contributing open technical research to the ecosystem. Derby provides an excellent base to access the Midlands' growing technology corridor while maintaining strong proximity to London's regulatory sector.

Conclusion

I seek endorsement not simply to continue a private technical project, but to develop AMTTP into UK-based compliance infrastructure with practical relevance to fintech, digital assets, and secure digital regulation. The evidence submitted in my application reflects a clear trajectory from independent research and development to public technical execution, intellectual property protection, and early external recognition.

I believe this demonstrates the innovation, promise, and likely future contribution expected under this route. I have the technical depth, the proven capacity for independent engineering, and the commercial foresight to become a leading technical contributor to the UK's digital technology sector.